

Solving $Ax = 0$: Pivot Variables, Special Solutions

Lesson 6 said what the do-nothing trades are. This lesson finds every one of them, for any menu of funds: simplify A to U and then to R , and read the rank, the pivot and free variables, and every solution off R .

PART 1 Index

One question runs through the lesson: **which trades change nothing?** We turn the matrix A into U , then into R , and read every answer off R .

1. The question, and a guess

Four funds, three rules. Two do-nothing trades are easy to guess. Are there more?

2. Step 1: from A to U

Clear below the pivots. Each new row explained in plain words; why this can't change the answer.

3. Step 2: from U to R

Make each pivot 1 and clear above it: the simplest form of the rules.

4. What the rank tells you

Count the pivots: how many rules carry information, how many funds bring something new.

5. Pivot variables and free variables

You pick the free amounts; R hands you the pivot amounts with no solving.

6. When does $Ax = 0$ have a non-zero solution?

Two cases: a free column (yes, infinitely many) or none (only $x = 0$). 2×2 and 3×3 examples.

7. Finding every solution from R

Special solutions read straight off R , then all their mixes. With a try-it box.

METHODS IN THIS LESSON

Row operations keep $N(A)$

Echelon form U

Reduced row echelon form R

Rank r

Pivot and free variables

$n - r$ rule

Special solutions

Nullspace matrix N

PART 2 Prerequisites

Click any link to review it, then come back.

- The nullspace (Lesson 6): the trades with $Ax = 0$, which change nothing.
- Elimination, 3×3 (Lesson 2): clearing below the pivots.
- Gauss–Jordan (Lesson 3): clearing above the pivots as well.

PART 3 Explanation

THE WHOLE LESSON IN ONE LINE

Simplify $A \rightarrow U \rightarrow R$. Then read off R : the **rank** (how many pivots), which amounts are **pivot** and which are **free**, **whether** any non-zero trade changes nothing, and **all** such trades.

1. The question, and a guess

In Lesson 6 a menu of three funds had one do-nothing trade, and we found it by hand. Real menus are bigger: many products, a few things you measure. So we need a method that works for **any** menu. Here is the menu we'll use for the whole lesson.

THE MENU: FOUR FUNDS, THREE RULES

Fund	What ₹1 lakh does to (budget, return, exposure)	Note
Nifty	(1, 12, 1)	12% return, beta 1
Nifty ETF	(1, 12, 1)	a twin of Nifty from another fund house
Bonds	(1, 7, 0)	7% return, beta 0
Balanced	(1, 9.5, 0.5)	half Nifty, half bonds

A trade is four amounts $x = (x_1, x_2, x_3, x_4)$: ₹ lakh into Nifty, ETF, bonds, balanced. Positive = buy, negative = sell.

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 12 & 12 & 7 & 9.5 \\ 1 & 1 & 0 & 0.5 \end{bmatrix} \begin{array}{l} \leftarrow \text{budget} \\ \leftarrow \text{return} \\ \leftarrow \text{exposure} \end{array}$$

The question: which trades x change nothing, so that $Ax = 0$? (That set is the nullspace $N(A)$.)

A guess

1 GUESS 1

The ETF is a copy of Nifty. **Buy ₹1 lakh ETF, sell ₹1 lakh Nifty:**
 $(-1, 1, 0, 0)$.

CHECK

Budget $-1 + 1 = 0$. Return $-12 + 12 = 0$. Exposure $-1 + 1 = 0$ ✓.

2 GUESS 2

Balanced is half Nifty, half bonds. **Buy ₹1 lakh balanced, sell ₹0.5 lakh each of Nifty and bonds:** $(-0.5, 0, -0.5, 1)$.

CHECK

Budget $-0.5 - 0.5 + 1 = 0$. Return $-6 - 3.5 + 9.5 = 0$. Exposure $-0.5 + 0 + 0.5 = 0$ ✓.

Guessing leaves questions open: **are there others we missed?** And what about a menu where nothing is obvious? The method below answers both, and it will produce these two guesses on its own.

2. Step 1: from A to U

Three rules about four funds are hard to read. The first job is to simplify them by elimination. First, why that's allowed.

Each row is a question about the trade

Row	The question it asks	Its formula
1	How much does the budget change?	$x_1 + x_2 + x_3 + x_4$
2	How much does the return change?	$12x_1 + 12x_2 + 7x_3 + 9.5x_4$
3	How much does the exposure change?	$x_1 + x_2 + 0x_3 + 0.5x_4$

A do-nothing trade answers **0 to every question**. Elimination replaces a question with a combination, like "return minus 12 × budget". A trade that answers 0 to both "return" and

"budget" answers $0 - 12 \times 0 = 0$ to the new one. And every step can be undone (return = new question $+12 \times$ budget), so no new trade can sneak in.

RULE · ROW OPERATIONS KEEP THE NULLSPACE

Subtracting a multiple of one row from another, swapping rows, or dividing a row by a non-zero number never changes which trades give all zeros. So $Ax = 0$, $Ux = 0$ and $Rx = 0$ have exactly the same solutions.

Clear below the pivots: three row operations

1 WHAT

Row 2 $- 12 \times$ row 1.

ENTRIES

$12 - 12(1) = 0$, $12 - 12(1) = 0$, $7 - 12(1) = -5$, $9.5 - 12(1) = -2.5$. New row 2: $(0, 0, -5, -2.5)$.

MEANING

"How much return **beyond** what the same money would earn in Nifty at 12%?" Nifty and its ETF: 0 extra. Bonds: 5 less. Balanced: 2.5 less.

2 WHAT

Row 3 $- 1 \times$ row 1.

ENTRIES

$1 - 1 = 0$, $1 - 1 = 0$, $0 - 1 = -1$, $0.5 - 1 = -0.5$. New row 3: $(0, 0, -1, -0.5)$.

MEANING

"Exposure minus budget": how much less exposure than an all-Nifty trade of the same size.

3 WHAT

Row 3 $- 0.2 \times$ row 2 (multiplier $= (-1)/(-5) = 0.2$).

ENTRIES

$0, 0, -1 - 0.2(-5) = -1 + 1 = 0$, $-0.5 - 0.2(-2.5) = -0.5 + 0.5 = 0$. New row 3: $(0, 0, 0, 0)$.

MEANING

Row 3 now asks a question whose answer is **always** 0. Exposure told us nothing that budget and return didn't already.

METHOD · ECHELON FORM U (THE STAIRCASE)

Eliminate column by column, clearing everything **below** each pivot. If a column has nothing left to use as a pivot, skip it and move right. The result U is a staircase: each row starts further right than the one above; all-zero rows sit at the bottom.

$$U = \begin{bmatrix} \boxed{1} & 1 & 1 & 1 \\ 0 & 0 & \boxed{-5} & -2.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + x_3 + x_4 = 0 \\ \text{row 2: } -5x_3 - 2.5x_4 = 0 \\ \text{row 3: } 0 = 0 \end{array}$$

The boxed entries are the **pivots**: the first non-zero entry of each row. Column 2 got no pivot because, after clearing Nifty, the ETF column was cleared too (it's the same column). Only **two** real rules are left.

IN ONE LINE

$A \rightarrow U$: clear below the pivots; the do-nothing trades don't change, and useless rules turn into zero rows.

3. Step 2: from U to R

U is simpler, but rule 1 still mixes all four funds, including bonds, which rule 2 is about. One more round of tidying makes each rule as simple as it can be.

1 WHAT

Make pivot 2 equal to 1: row 2 $\div (-5)$.

ENTRIES

0, 0, $-5/(-5) = 1$, $-2.5/(-5) = 0.5$. New row 2: (0, 0, 1, 0.5).

MEANING

$x_3 + 0.5x_4 = 0$: "bonds amount + half the balanced amount = 0".

2 WHAT

Clear **above** pivot 2: row 1 $- 1 \times$ row 2.

ENTRIES

$1 - 0 = 1$, $1 - 0 = 1$, $1 - 1 = 0$, $1 - 0.5 = 0.5$. New row 1: (1, 1, 0, 0.5).

MEANING

$x_1 + x_2 + 0.5x_4 = 0$: bonds are gone from rule 1. (Pivot 1 is already 1, so nothing to divide.)

METHOD · REDUCED ROW ECHELON FORM R

From U : divide each pivot row by its pivot (every pivot becomes 1), then clear the entries **above** each pivot (Gauss-Jordan, Lesson 3). The result $R = \text{rref}(A)$: every pivot is 1 and is the only non-zero entry in its column.

$$R = \begin{bmatrix} \boxed{1} & 1 & 0 & 0.5 \\ 0 & 0 & \boxed{1} & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix} \quad \begin{array}{l} \text{row 1: } x_1 + x_2 + 0.5x_4 = 0 \\ \text{row 2: } x_3 + 0.5x_4 = 0 \\ \text{row 3: } 0 = 0 \end{array}$$

Look at what's special: each rule now contains **exactly one** of Nifty and bonds, plus only ETF and balanced. Rule 1 is about Nifty; rule 2 is about bonds. That's what makes everything in the next topics a matter of reading.

IN ONE LINE

$U \rightarrow R$: pivots become 1 and everything above them is cleared; each rule now involves one pivot amount only.

4. What the rank tells you

Count the pivots in R (the same as in U): there are **two**, in columns 1 and 3.

DEFINITION · RANK R = THE NUMBER OF PIVOTS

The rank of A is how many pivots elimination finds. Here $r = 2$.

That single number says three things about the menu:

The rank tells you...	Here	Why
how many rules really carry information	2 of 3	row 3 became all zeros: exposure was already decided by budget and return
how many funds bring something new	2 of 4	Nifty and bonds; the ETF repeats Nifty, balanced mixes Nifty and bonds
how many amounts are left free	4 – 2 = 2	each pivot rule pins down one amount; the rest are not pinned

The rank can never be more than the number of rows ($m = 3$) or the number of columns ($n = 4$), since each pivot needs its own row and its own column.

IN ONE LINE

Rank = pivots = the rules that matter = the funds that bring something new.

5. Pivot variables and free variables

The rank said two amounts are pinned down and two are free. R says **which**.

RULE · PIVOT VARIABLES AND FREE VARIABLES

A column with a pivot is a **pivot column**; its amount is a **pivot variable** (here Nifty x_1 , bonds x_3). A column with no pivot is a **free column**; its amount is a **free variable** (here ETF x_2 , balanced x_4). You choose the free variables; R then gives the pivot variables.

Try it with numbers: you pick, R decides

Say you decide: buy ₹3 lakh ETF ($x_2 = 3$) and ₹2 lakh balanced ($x_4 = 2$). Each row of R now hands you one pivot amount directly:

1 ROW 2 OF R

$$x_3 + 0.5(2) = 0 \Rightarrow x_3 = -1. \text{ Sell ₹1 lakh bonds.}$$

ROW 1 OF R

$$x_1 + 3 + 0.5(2) = 0 \Rightarrow x_1 = -4. \text{ Sell ₹4 lakh Nifty.}$$

2 THE TRADE

$$(-4, 3, -1, 2).$$

CHECK IN A

$$\text{Budget } -4 + 3 - 1 + 2 = 0 \checkmark. \text{ Return } -48 + 36 - 7 + 19 = 0 \checkmark.$$

$$\text{Exposure } -4 + 3 + 0 + 1 = 0 \checkmark. \text{ It changes nothing.}$$

Whatever free amounts you pick, the rows of R give, with no solving at all:

$$\text{Nifty } x_1 = -x_2 - 0.5 x_4$$

$$\text{bonds } x_3 = -0.5 x_4$$

Why the ETF and balanced are the free ones

Each row of R starts at a pivot, and that row is the rule that decides that pivot's amount: row 1 decides Nifty, row 2 decides bonds. **No row starts at the ETF or balanced**, so nothing ever decides them: they're yours to pick.

In money terms, the free funds are exactly the ones that **add nothing new**, so the pivot funds can always undo them:

You buy	Why it can be undone	To undo it
₹3 lakh ETF	it is a copy of Nifty	sell ₹3 lakh Nifty
₹2 lakh balanced	it is half Nifty, half bonds	sell ₹1 lakh Nifty and ₹1 lakh bonds
Both		sell ₹4 lakh Nifty and ₹1 lakh bonds

That's the trade $(-4, 3, -1, 2)$ that R produced. The algebra and the common sense agree.

▼ For the curious: could Nifty be free and the ETF be the pivot instead?

Yes: the twins are interchangeable, so you could choose the Nifty amount and let the ETF be forced. Elimination avoids that choice by reading **left to right**: a column gets a pivot only if it brings something new compared with the columns *before* it. Nifty comes first, so Nifty gets the pivot. Either way you end up with the same set of do-nothing trades.

IN ONE LINE

Pivot variables are decided by the rows of R ; free variables are the amounts no row decides, and you choose them.

6. When does $Ax = 0$ have a non-zero solution?

"Do nothing", $x = 0$, always solves $Ax = 0$. The real question is whether some **non-zero** trade also changes nothing. Topic 5 gives the answer straight away: it all depends on whether there is a free column.

RULE · FREE COLUMNS DECIDE IT ($N - R$ OF THEM)

There are $n - r$ free columns. **Case 1:** $n - r > 0$ (some free column): pick a non-zero free amount and R completes it into a non-zero do-nothing trade, so there are **infinitely many**. **Case 2:** $n - r = 0$ (every column has a pivot): every amount is decided by R , and with all right sides 0 they are all 0, so **only** $x = 0$. Since $r \leq m$, more funds than rules ($n > m$) always means Case 1.

2 × 2: Nifty and its twin, budget and return

1 $A \rightarrow U$

$U \rightarrow R$

$\begin{bmatrix} 1 & 1 \\ 12 & 12 \end{bmatrix}$. Row 2 $- 12 \times$ row 1: $12 - 12 = 0$, $12 - 12 = 0$. $U = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$.

Pivot is already 1 with nothing above: $R = U$.

READ

Rank 1, ETF free. **Case 1:** $\text{ETF} = 1$ gives $\text{Nifty} = -1$: the trade $(-1, 1)$, and all its multiples.

2 × 2: Nifty and bonds, budget and return

1 A → U

$\begin{bmatrix} 1 & 1 \\ 12 & 7 \end{bmatrix}$. Row 2 $- 12 \times$ row 1: $12 - 12 = 0, 7 - 12 = -5$. $U = \begin{bmatrix} 1 & 1 \\ 0 & -5 \end{bmatrix}$

U → R

Row 2 $\div (-5)$: $(0, 1)$. Row 1 $-$ row 2: $(1, 0)$. $R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

READ

Rank 2 = 2 columns, nothing free. **Case 2:** R says $x_1 = 0$ and $x_2 = 0$. Only "do nothing".

3 × 3: Nifty, bonds, balanced

1 A → U

The same three row operations as topic 2, without the ETF column:
 $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & -2.5 \\ 0 & 0 & 0 \end{bmatrix}$.

U → R

Row 2 $\div (-5)$: $(0, 1, 0.5)$. Row 1 $-$ row 2: $(1, 0, 0.5)$. $R = \begin{bmatrix} 1 & 0 & 0.5 \\ 0 & 1 & 0.5 \\ 0 & 0 & 0 \end{bmatrix}$.

READ

Rank 2, balanced free. **Case 1:** balanced = 1 gives $\text{Nifty} = -0.5$, bonds = -0.5 : $(-0.5, -0.5, 1)$. ($\times -2$ gives Lesson 6's $(1, 1, -2)$.)

3 × 3: Nifty, bonds, midcap

1 A → U

Row 2 $- 12 \times$ row 1: $(0, -5, 15 - 12 = 3)$. Row 3 $-$ row 1: $(0, -1, 1.2 - 1 = 0.2)$. Row 3 $- 0.2 \times$ row 2: $(0, 0, 0.2 - 0.6 = -0.4)$. $U = \begin{bmatrix} 1 & 1 & 1 \\ 0 & -5 & 3 \\ 0 & 0 & -0.4 \end{bmatrix}$.

U → R

Row 3 $\div (-0.4)$: $(0, 0, 1)$. Row 2 $- 3 \times$ row 3: $(0, -5, 0)$; then $\div (-5)$: $(0, 1, 0)$. Row 1 $-$ row 3 $-$ row 2: $(1, 0, 0)$. $R = I$.

READ

Rank 3 = 3 columns, nothing free. **Case 2:** only $x = 0$. Midcap brings something new, so no fund can be undone by the others.

Menu	m rules	n funds	rank r	free $n - r$	Non-zero do-nothing trade?
Nifty, twin	2	2	1	1	yes (Case 1)
Nifty, bonds	2	2	2	0	no (Case 2)
Nifty, bonds, balanced	3	3	2	1	yes (Case 1)
Nifty, bonds, midcap	3	3	3	0	no (Case 2)
Nifty, ETF, bonds, balanced	3	4	2	2	yes (Case 1, guaranteed since $4 > 3$)

WHY A DESK CARES

Three things you measure and four or more products: a way to switch between products that changes nothing you measure is **guaranteed** to exist. That's where a desk looks for the cheaper route, and also where risks you *don't* measure can slip in.

IN ONE LINE

A free column $\Rightarrow$ non-zero do-nothing trades exist (infinitely many); no free column $\Rightarrow$ only $x = 0$.

7. Finding every solution from R

We know **when** non-zero solutions exist. Now we find **all** of them, by choosing the free amounts in the simplest way: one at a time.

METHOD · SPECIAL SOLUTIONS, READ OFF R

For each free variable in turn: set it to 1 and the other free variables to 0. Each pivot variable is then **minus** the number in that free column of R , in the pivot's row. One free column gives one special solution.

$$R = \begin{bmatrix} 1 & 1 & 0 & 0.5 \\ 0 & 0 & 1 & 0.5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

ETF column (green): (1, 0)
balanced column (pink): (0.5, 0.5)

1 ETF = 1

RESULT

balanced = 0. ETF column of R is $(1, 0)$, so Nifty = -1 , bonds = 0.
 $s_1 = (-1, 1, 0, 0)$. That's **guess 1**: swap Nifty for its twin.

2 BALANCED = 1

RESULT

ETF = 0. Balanced column of R is $(0.5, 0.5)$, so Nifty = -0.5 , bonds = -0.5 .
 $s_2 = (-0.5, 0, -0.5, 1)$. That's **guess 2**: build the balanced fund yourself.

Check in A	Budget 1, 1, 1, 1	Return 12, 12, 7, 9.5	Exposure 1, 1, 0, 0.5
$s_1 = (-1, 1, 0, 0)$	$-1 + 1 + 0 + 0 = 0 \checkmark$	$-12 + 12 + 0 + 0 = 0 \checkmark$	$-1 + 1 + 0 + 0 = 0 \checkmark$
$s_2 = (-0.5, 0, -0.5, 1)$	$-0.5 + 0 - 0.5 + 1 = 0 \checkmark$	$-6 + 0 - 3.5 + 9.5 = 0 \checkmark$	$-0.5 + 0 + 0 + 0.5 = 0 \checkmark$

The method found our two guesses without any guessing.

Every do-nothing trade is a mix of these

1 ANY CHOICE

SPLIT IT

Choose ETF = c and balanced = d . R gives Nifty = $-c - 0.5d$ and bonds = $-0.5d$.

$$\begin{bmatrix} -c - 0.5d \\ c \\ -0.5d \\ d \end{bmatrix} = c \begin{bmatrix} -1 \\ 1 \\ 0 \\ 0 \end{bmatrix} + d \begin{bmatrix} -0.5 \\ 0 \\ -0.5 \\ 1 \end{bmatrix} = c s_1 + d s_2.$$

2 NOTHING ELSE

EXAMPLE

Every do-nothing trade is pinned down by its free amounts. If both are 0, R makes the pivot amounts 0 too. So there is no hidden third kind: the answer to "are there others?" is **no**.

The trade $(-3, 2, -1, 2)$ has ETF 2 and balanced 2, so it is $2s_1 + 2s_2 = (-2 - 1, 2, -1, 2) \checkmark$.

Two free choices give two directions, so $N(A)$ is a **plane** inside $\mathbb{R}^4$ (the space of four-fund trades).

Try it: choose the free amounts

ETF amount $x_2 = c$

Balanced amount $x_4 = d$

Free (you chose): ETF $x_2 = 2$, balanced $x_4 = 2$.

Forced: Nifty $x_1 = -c - 0.5d = -3$, bonds $x_3 = -0.5d = -1$.

Trade $= 2s_1 + 2s_2 = (-3, 2, -1, 2)$ (₹ lakh in Nifty, ETF, bonds, balanced).

Budget: $1(-3) + 1(2) + 1(-1) + 1(2) = 0$ ✓

Return: $12(-3) + 12(2) + 7(-1) + 9.5(2) = 0$ ✓

Exposure: $1(-3) + 1(2) + 0(-1) + 0.5(2) = 0$ ✓

Nothing changes: every rule stays at 0, whatever free amounts you pick.

All of them in one matrix

METHOD · NULLSPACE MATRIX N

Put the special solutions side by side as columns. Then $N \begin{bmatrix} c \\ d \end{bmatrix} = cs_1 + ds_2$ lists every do-nothing trade, and $RN = 0$ (and $AN = 0$).

$$N = \begin{bmatrix} -1 & -0.5 \\ 1 & 0 \\ 0 & -0.5 \\ 0 & 1 \end{bmatrix} \begin{array}{l} \leftarrow \text{Nifty} \\ \leftarrow \text{ETF} \\ \leftarrow \text{bonds} \\ \leftarrow \text{balanced} \end{array}$$

Check $RN = 0$	$s_1 = (-1, 1, 0, 0)$	$s_2 = (-0.5, 0, -0.5, 1)$
Row 1 of R : $(1, 1, 0, 0.5)$	$-1 + 1 + 0 + 0 = 0$ ✓	$-0.5 + 0 + 0 + 0.5 = 0$ ✓
Row 2 of R : $(0, 0, 1, 0.5)$	$0 + 0 + 0 + 0 = 0$ ✓	$0 + 0 - 0.5 + 0.5 = 0$ ✓
Row 3 of R : $(0, 0, 0, 0)$	0 ✓	0 ✓

▼ For the curious: Strang's compact form $R = [I \ F]$ and $N = [-F; I]$

List the pivot columns first (Nifty, bonds) and the free columns after (ETF, balanced). The pivot part of R is then the identity I and the free part is $F = \begin{bmatrix} 1 & 0.5 \\ 0 & 0.5 \end{bmatrix}$. The reading rule says pivot amounts $= -F$, free amounts $= I$, so in that order $N = \begin{bmatrix} -F \\ I \end{bmatrix}$ and $RN = I(-F) + FI = 0$.

MIT's lecture matrix $\begin{bmatrix} 1 & 2 & 2 & 2 \\ 2 & 4 & 6 & 8 \\ 3 & 6 & 8 & 10 \end{bmatrix}$ has the same shape as our menu: $R = \begin{bmatrix} 1 & 2 & 0 & -2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$, pivots in columns 1 and 3, special solutions $(-2, 1, 0, 0)$ and $(2, 0, -2, 1)$.

▼ Check yourself (MIT recitation): all solutions of $x - 5y + 2z = 0$.

$R = [1 \ -5 \ 2]$ already (one row, pivot 1). Rank 1, y and z free. Reading rule: $y = 1$ gives $x = -(-5) = 5$, so $(5, 1, 0)$; $z = 1$ gives $x = -2$, so $(-2, 0, 1)$. All solutions: $c(5, 1, 0) + d(-2, 0, 1)$, a plane.

▼ Check yourself (MIT problem 7.1): find R , the rank and the special solutions of $\begin{bmatrix} 1 & 5 & 7 & 9 \\ 0 & 4 & 1 & 7 \\ 2 & -2 & 11 & -3 \end{bmatrix}$.

$A \rightarrow U$: row 3 $- 2 \times$ row 1 $= (0, -12, -3, -21)$; row 3 $+ 3 \times$ row 2 $= (0, 0, 0, 0)$. $U \rightarrow R$: row 2 $\div 4 = (0, 1, \frac{1}{4}, \frac{7}{4})$; row 1 $- 5 \times$ row 2 $= (1, 0, \frac{23}{4}, \frac{1}{4})$. Rank 2, free columns 3 and 4. Reading rule: $(-\frac{23}{4}, -\frac{1}{4}, 1, 0)$ and $(-\frac{1}{4}, -\frac{7}{4}, 0, 1)$. Try it in the lab on the Examples tab.

IN ONE LINE

One free amount = 1, the rest 0, pivot amounts = $-$ (that column of R): the special solutions; all their mixes are every do-nothing trade.